Part 19 — Protein products, and the theory the problems had outrun

Two things, and the second is the more important. First, protein products — secretion, inclusion bodies, refolding, glycosylation — which the whole module had treated as an aside because it was built around small molecules and polymers. Second, an audit I should have run earlier: whether the module actually teaches the theory my own problems assume. It did not, in five places.

The audit, and what it found

I searched all fifteen module parts for the theory behind every problem I wrote. Most was there. These were not:

Theory a problem assumes	Problem	In the module before
Analysis of variance and interaction testing	X13	**Absent**
Variance components and nested replication	X15	**Absent**
Equivalence testing	X16	**Absent**
Gas holdup and impeller flooding	X11	**Absent**
Dissolved CO_2 and its pH consequences	X9	**Absent**
Response-surface designs	X14	Thin — named, never derived
Diafiltration and sieving coefficients	Z4	Thin
Protein secretion, refolding	PR1–PR8	Thin — scattered mentions

That is a fair criticism of how this was built: I wrote problems faster than the chapters that justify them, so for eight problems the worked solution was the only explanation available. Chapters T1 to T5 fix it, and PR1 to PR8 are the new protein problems — taking the set to **125**.

Chapter T1 · extends weeks 8, 9 and 30

Protein products I: expression, folding and inclusion bodies

A protein product is the one case where the thing you are making is also the thing doing the making. Everything that follows comes from that: the proteome budget, the folding competition, and the fact that a failed product is not lost carbon but a precipitate.

What makes a protein product different

For a small molecule, the pathway is the catalyst and the product is separate. For a protein, the ribosome makes the product directly, so **yield is set by the proteome budget, not by a carbon ceiling**. Part 14 chapter B9 gives the arithmetic: a cell holds about 0.5 g of protein per gram dry weight, and a heterologous product competes with that directly. At 30 gDW/L a product at 5 g/L is a third of one proteome-equivalent, and at 20 g/L it exceeds the cell's own protein content — possible only because the product leaves the cell while the cells keep growing, which is the first argument for secretion.

The second difference is that a protein can be made and still be worthless. A small molecule is right or absent; a protein can be **expressed, insoluble, misfolded, truncated, proteolysed, or correctly folded and wrongly modified**. So the quantity to measure is never "how much protein" but "how much active, correctly folded protein", and part 7 chapter 22's warning that a band is not an activity is the governing rule.

The folding competition, quantitatively

A newly synthesised chain either folds or aggregates, and the two differ in order. Folding is first order in the unfolded chain; aggregation requires two chains to meet, so it is second order. With rate constants k_f and k_a and an unfolded pool [U], the soluble fraction is approximately

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soluble fraction ≈ k_f /(k_f + k_a[U]) = 1/(1 + (k_a/k_f)[U])
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k_a/k_f	[U] = 0.1 µM	1 µM	5 µM
0.5	95%	67%	29%
2.0	83%	33%	9%
10	50%	9%	2%

The asymmetry in order is the whole of inclusion-body engineering. Because aggregation depends on [U] and folding does not, **anything that lowers the unfolded pool raises the soluble fraction** — and the unfolded pool is set by the synthesis rate. That is why every one of the standard remedies works:

Intervention	What it does to the arithmetic	Cost
Lower the induction level	Halves [U], roughly doubles the soluble fraction at $k_a/k_f = 2$	Less total protein — the trade is soluble yield against total
Drop 37 °C to 25 or 16 °C	Slows synthesis several-fold and slows aggregation more than folding	Slower growth, longer run
Co-express chaperones	Raises k_f and holds chains in a foldable state	Proteome, computed below
Fusion partners (MBP, SUMO, thioredoxin)	Raise intrinsic solubility; SUMO is cleavable without a scar	Tag removal is a step, with its own yield
Periplasmic targeting	Oxidising environment lets disulfides form; different chaperones	Translocation capacity, and a volume ceiling (T2)

Chaperones illustrate how to decide. Co-expression costs proteome and buys solubility, so the net gain is the solubility factor times the burden penalty of part 14 chapter B9, $\mu/\mu_0 = 1 - \varphi/0.55$. At a 10% chaperone load the penalty is 0.82, so **the break-even solubility gain is 1.22-fold** — below that you are worse off. A threefold gain at a 10% load nets 2.45-fold; the same gain at a 2% load nets 2.89.

When inclusion bodies are the plan

They are not always a failure. Inclusion bodies are dense, nearly pure, protease-resistant, and easy to separate by

centrifugation — so for a robust protein with no disulfides and a known refolding route, the deliberate route is **express into inclusion bodies at maximum rate, wash, solubilise, refold**, and the overall yield can beat a soluble route that expresses weakly. The decision turns on whether the protein refolds, which is T1's companion question and chapter T2's arithmetic.

What makes them unattractive: a protein with several disulfides, a protein that needs a cofactor inserted co-translationally, anything where refolding yield is below about 30%, and any case where the regulatory burden of demonstrating correct refolding is heavier than the fermentation saving.

Proteome limit	`product per gDW against 0.5 g/gDW; 20 g/L at 30 gDW/L exceeds one proteome-equivalent`
Folding competition	`soluble ≈ 1/(1 + (`k_a/k_f`)[U]); aggregation is second order, folding first`
The lever	`lower [U] — induction level, temperature, promoter strength`
Chaperone break-even	`gain > 1/(1 − φ/0.55); 1.22-fold at a 10% load`
Inclusion bodies as a plan	`viable when refolding yield is high and disulfides are few`

What to open a book for. A review of recombinant protein production in *E. coli* for the solubility interventions and their published success rates; Baneyx & Mujacic, Nat. Biotechnol. 22:1399 (2004) for folding and inclusion bodies, read for the mechanisms rather than the protocols.

Chapter T2 · extends weeks 58, 59 and 49

Protein products II: secretion, refolding, glycosylation and formulation

Secretion removes the intracellular ceiling and replaces it with a translocation ceiling. Refolding trades concentration against yield. Glycosylation decides whether the molecule is a drug or a liability.

Secretion, and the three ceilings it moves

A secreted product escapes the volume ceiling of part 6 chapter 59 entirely — there is no intracellular pool to saturate. What replaces it is **translocation capacity**. In the Sec pathway an unfolded chain is threaded through a translocon, so the flux per cell is the number of translocons times the threading rate divided by the chain length. At roughly 100 translocons per cell† and 20 residues per second, a 300-residue protein can leave at about 24,000 molecules per cell per hour, which at 10^9 cells per millilitre is of order **1 g·L⁻¹·h⁻¹ at full occupancy** — enough for grams per litre over a long run, and only just, which is why secretion titers plateau where they do.

Three routes, with different constraints. **Sec** takes unfolded chains and is the high-capacity default. **Tat** takes *folded* chains, which is the only option for a protein with a co-translationally inserted cofactor, and has far lower capacity. In yeasts the route is the **ER and Golgi**, which adds glycosylation and quality control: misfolded chains are retro-translocated and degraded, so secretion yield depends on folding in a compartment you cannot easily see.

For an *E. coli* product, the usual compromise is the **periplasm**: oxidising, so disulfides form, with DsbA and DsbC to catalyse and reshuffle them. The ceiling is volumetric — the periplasm is roughly 15%† of cell volume, so at 100 gDCW/L and 2 mL/gDCW it is 30 mL per litre of broth, and a protein at 5 mM there is **5 g/L of broth**. That is a real ceiling and it is why true secretion into the medium, where *E. coli* is poor and *Bacillus* and *K. phaffii* are good, is worth a host change.

Refolding: the dilution trade

Solubilised inclusion bodies are refolded by removing the denaturant, and the same competition as T1 applies with the same orders: folding first, aggregation second. So the yield falls with concentration:

Refolding concentration	Yield	Protein recovered per litre
0.01 g/L	93%	0.009 g
0.1 g/L	56%	0.056 g
1.0 g/L	11%	0.111 g

Note what the third column does: **protein per litre of vessel rises monotonically while yield collapses**, so there is no interior optimum and the choice is purely vessel volume against material loss. The engineering answer is to decouple the two by **pulse or fed-batch refolding**: add the denatured protein in portions so the local concentration stays low while the total is high. Adding 1 g in twenty pulses at 0.05 g/L each recovers **0.714 g (71%)** against 0.111 g (11%) in one shot — a six-fold improvement from the addition schedule alone. The same logic underlies continuous dilution refolding and size-exclusion refolding.

Glycosylation

N-linked glycans are attached at Asn-X-Ser/Thr sequons and matter for three reasons: **serum half-life**, since poorly sialylated glycoproteins are cleared quickly; **immunogenicity**, since non-human glycans — yeast high mannose, plant xylose and core fucose, the mammalian Neu5Gc and α-gal epitopes — provoke responses; and **activity**, since glycans at the Fc region modulate effector function.

Two numbers worth holding. **Occupancy is never complete**: at 85% per site, a protein with two sites is 72% fully occupied, with three sites 61%, with five sites 44% — so a multi-sequon protein is intrinsically a mixture, and "the product" is a distribution you must specify. And yeast **hypermannosylation** is a mass problem, not just an immunogenicity one: a human core glycan of ten mannose adds about 1.6 kDa per site, while a yeast outer chain of 50 to 150 adds 8 to 24 kDa — enough to change the molecule's chromatographic behaviour and its apparent mass.

Which host follows from that. **Bacteria do not glycosylate**, so for a glycoprotein they are out unless the glycan is dispensable. **Yeasts** glycosylate wrongly unless humanised — deleting OCH1 and installing a human pathway, each step costing growth. **CHO cells** glycosylate acceptably and are slow and expensive. That is the table in part 6 chapter 61, and glycosylation is the row that usually decides it.

Formulation, briefly, because it decides the specification

Aggregation continues after purification, second order in concentration, so $1/C - 1/C_0 = kt$. At $k = 10^{-3}$ L·g^{-1}·day^{-1} a 1 g/L formulation loses 10% in 111 days and a 10 g/L one loses it in **11 days** — tenfold concentration costs tenfold stability. This is why formulation screens are run at the final concentration and why a high-concentration product needs excipients. For an engineer it matters because **the fermentation titer and the formulation concentration are different numbers**, and a process optimised for the first can be inconvenient for the second.

Secretion capacity	translocons × rate ÷ length; ~1 g·L^{-1}·h^{-1} at full occupancy†
Periplasmic ceiling	~15% of cell volume → about 5 g/L of broth at 100 gDCW/L and 5 mM
Refolding	yield falls with concentration; pulsed addition gives 71% against 11%
Glycan occupancy	0.85^n fully occupied; 44% at five sites
Storage	second order: tenfold concentration, tenfold faster loss

† Translocon count, threading rate and periplasmic volume fraction are recalled from memory, not computed — verify them and record them on sheet 10 of the lookup workbook. Everything computed from them is arithmetic and is reliable given the inputs.

What to open a book for. A review of secretory protein production in *K. phaffii* and in *Bacillus*; the glycoengineering literature for humanised yeast; and, for refolding, a methods review on dilution and on-column refolding — the pulse-addition result above is the single most useful idea in it.

Chapter T3 · extends week 17

The statistics the problems assumed: ANOVA, response surfaces, variance components, equivalence

Part 2 chapter 17 taught replication, power and multiplicity. These four tools are what problems X13 to X16 actually use, and none of them was in the module.

Analysis of variance, and why it exists

With two factors you could run a t-test per comparison, and the multiplicity arithmetic of part 2 chapter 17 then eats you. ANOVA instead **partitions the total variation** into named parts and tests each against the residual. For two factors A and B with n replicates per cell:

```
SS_total = SS_A + SS_B + SS_AB + SS_error,     F = MS_effect / MS_error
```

Each mean square is its sum of squares divided by its degrees of freedom, and the F statistic is compared with the F distribution on those degrees of freedom. What matters for design is the **interaction term, SS_{AB}**: it is the variation that neither factor explains alone, and it exists only if you ran the combinations. One-factor-at-a-time generates no cell in which both factors are changed together, so it cannot estimate the interaction at any sample size — that is a structural blindness, not an efficiency problem, and it is what problem X13 demonstrates with a p = 0.0061 interaction.

Two cautions. A significant F says the factor explains variation, not that the effect is large: report the effect size and the fraction of variance explained alongside. And ANOVA assumes roughly equal variances and independent residuals, which a nested design (chapter T3's third section) violates.

Response-surface designs

A factorial finds which factors matter; a response surface finds *where the optimum is*, by fitting a quadratic:

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y = β₀ + Σβᵢxᵢ + Σβᵢᵢxᵢ² + Σβᵢⱼxᵢxⱼ
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A **central composite design** supplies the runs: the factorial corners give the interaction terms, axial points at ±α give the quadratic terms, and replicated centre points give an estimate of pure error. Three things then come out that a grid cannot give. The **stationary point**, from solving ∇y = 0. Its **character**, from the eigenvalues of the matrix of second derivatives — all negative is a maximum, mixed signs a saddle, and a saddle mistaken for an optimum sends the next round in a useless direction. And a **lack-of-fit test**, comparing the residual variation with the pure error from the centre replicates: if lack of fit is significant, the quadratic is the wrong model and the optimum it reports is not trustworthy.

Variance components and nested replication

When measurements are nested — injections within flasks within strains — the variance decomposes:

```
Var(total) = σ²_strain + σ²_flask + σ²_analytical
```

and each level is estimated from the mean squares of a nested ANOVA. The standard error of a strain mean with f flasks and a injections each is $\sqrt{(\sigma^2_{flask}/f + \sigma^2_{analytical}/(f \cdot a))}$, which is why **the level you replicate must be the level that varies**. Injections are *pseudo-replicates*: they estimate analytical variance, and if that is 2% of the total — as in problem X15 — they buy almost nothing. The general rule is to compute the components before designing the next experiment, because the cheapest replication is almost never the useful one.

Equivalence testing

A non-significant difference is not evidence of equivalence; it is usually evidence of a small experiment. To claim equivalence you invert the hypotheses. Choose an **equivalence margin δ** — the largest difference that would not change a decision — and run **two one-sided tests**: that the difference exceeds −δ, and that it is below +δ. Both must reject for equivalence to be claimed, which is identical to requiring the **confidence interval to lie entirely inside ±δ**.

The margin is a process decision made *before* the data, and the defensible choice is usually the run-to-run variability of the process, since a difference below that is unobservable in manufacturing. Problem X16 shows the consequence: the

same five-replicate comparison is "not equivalent" at ±0.5 g/L and "equivalent" at ±1.0, and the conventional t-test says nothing either way.

ANOVA	partition SS; $F = MS_{effect}/MS_{error}$; interaction needs the combinations
Response surface	quadratic fit; stationary point; eigenvalues classify it; centre points give lack of fit
Variance components	$SE = \sqrt{(\sigma^2_{flask}/f + \sigma^2_{analytical}/(f \cdot a))}$; replicate what varies
Equivalence	TOST at a pre-registered margin; CI inside ±δ

What to open a book for. Box, Hunter & Hunter, *Statistics for Experimenters*, for factorials and response surfaces — it is the standard and it is readable. For equivalence testing, a short methods paper on TOST is enough.

Chapter T4 · extends week 44

The reactor physics the problems assumed: heat, shear, holdup and dissolved CO_2

Part 11 chapter 44 did mixing, oxygen transfer and feeding. These four decide whether a vessel that can supply oxygen can also be cooled, aerated and kept at pH.

Heat: the same calculation as oxygen

Aerobic metabolism releases about **460 kJ per mole of oxygen consumed**†, nearly independent of substrate — which is the useful fact, because it means the oxygen calculation you already did is also the heat calculation. Heat duty is OUR × volume × 460 kJ/mol, and the area to remove it is Q/(U·ΔT).

Two consequences the problems computed. A glucose fed-batch at 40 gDW/L and 4 mmol·gDW^{-1}·h^{-1} releases 2.04 MW in 100 m³ and needs 408 m² against roughly 110 m² of jacket, so **internal cooling is needed above about 11 gDW/L**. And a methanol process at 30 gDW/L releases 4.28 MW — twice as much for less biomass, because a reduced substrate demands more oxygen per gram. Heat, not oxygen, is often what sizes the vessel, and it is a capital decision taken at design time.

Shear: a question of length scales

Turbulent energy cascades to ever smaller eddies until viscosity dissipates it, at the Kolmogorov scale $\lambda = (\nu^3/\varepsilon)^{1/4}$, with $\varepsilon = (P/V)/\rho$. Because the dependence is a fourth root, **λ falls very slowly with power** — 56 µm at 100 W/m³, 32 µm at 1,000, 15 µm at 20,000.

Compare the scale with the particle. A cell much smaller than λ is carried by the eddy rather than deformed by it, so **a 1 µm bacterium is not damaged by shear in a stirred tank at any realistic power**, and reports to the contrary usually concern the air–liquid interface or foam. A 15 µm animal cell is comparable to λ at high power, which is why those cultures run at tens of W/m³. A 100 µm floc, pellet or immobilisation bead is larger than λ everywhere and will erode — so pellet processes are limited by impeller design and tip speed, not by mean power. Local dissipation near the impeller is 20 to 50 times the mean, dividing λ by about 2.3, and cells circulate through that zone repeatedly.

Gas holdup and flooding

Holdup is the volume fraction of gas in the aerated broth, rising roughly as the superficial gas velocity to the power 0.7. It matters twice: it is working volume you no longer have — 12% at 0.1 m/s — and it is where the interfacial area for transfer comes from, so the same gas that costs volume buys k_La.

Flooding is the limit. Above a flow number Q/(ND³) of about 0.25 the impeller can no longer disperse the gas; bubbles coalesce and rise as a column past the blades, and k_La *stops rising with gas flow*. CO_2 stripping stops improving at the same point, because it depends on the same interfacial area. Flooding arrives sooner at low stirrer speed — exactly when a control system is trying to save power — so the practical rule is to check the flow number before increasing aeration, and to accept that beyond it the only levers are oxygen enrichment, head pressure and lower demand.

Dissolved CO_2, the variable nobody controls

At RQ = 1 the vessel must strip CO_2 as fast as it supplies oxygen, and the same k_La governs both, so the excess dissolved CO_2 at steady state is **CER/k_La**. At 250 mmol·L^{-1}·h^{-1} and k_La 100 h^{-1} that is 2.5 mM — about 0.1 atm equivalent — and it does three things. It **acidifies**, through the CO_2/bicarbonate equilibrium at $pK_1 \approx 6.1$, so at pH 7 each 1.5 mM of dissolved CO_2 carries about 12 mM of bicarbonate and consumes base. It **inhibits** some organisms directly. And it **rises with everything you do to help oxygen**: head pressure multiplies dissolved CO_2 and O_2 in the same ratio, so a vessel pressurised for oxygen transfer gives its cells proportionally more CO_2.

This is the classic scale-up variable that bench vessels hide, because a small vessel with a short liquid path strips CO_2 easily. Watch base consumption and the pH control output, which move before growth does.

Heat	`Q = OUR·V·460 kJ/mol; area = Q/(U·ΔT); jacket runs out near 11 gDW/L`
Shear	`λ = (ν³/ε)1/4; bacteria safe, 100 µm pellets not`
Holdup, flooding	`holdup ∝ v_s0.7; flooding above Q/(ND³) ≈ 0.25`
Dissolved CO_2	`excess = CER/k_La; pK₁ 6.1; rises with head pressure`

What to open a book for. Doran, the mass-transfer and heat-transfer chapters, worked rather than read; a mixing handbook for the power and flooding correlations, where the exponents are tabulated by impeller type.

Chapter T5 · extends chapter S1

The downstream theory the problems assumed: breakthrough, diafiltration, disruption

Chapter S1 gave the train and the costs. These are the three unit operations whose governing equations the problems used without deriving them.

Breakthrough and dynamic capacity

Loading a column is a moving concentration front. Ahead of it the resin is empty; behind it the resin is saturated; between them is a **mass-transfer zone** whose length is set by how fast the solute can diffuse into the bead relative to how fast the liquid moves past it. Breakthrough begins when the leading edge of that zone reaches the column outlet, which is before the bed is full — so **dynamic binding capacity is always below static capacity**, and the deficit is the unsaturated part of the zone.

Two design consequences follow. The unused length is roughly independent of bed height, so a **taller bed wastes a smaller fraction** — hence tall, narrow capture columns. And since the zone narrows as residence time rises, dynamic capacity increases with contact time, which is why DBC must always be quoted with its flow rate. Breakthrough also defines the yield: a capacity measured at 10% breakthrough means 10% of the feed leaves unbound at the end of loading, so capacity and loss are two readings of one curve.

Diafiltration: two sieving coefficients

Washing a small impurity out of a retained product by adding buffer at the permeate rate gives a first-order washout in **diavolumes**, N, the buffer volume divided by the retentate volume. For a species with sieving coefficient S — the ratio of its permeate concentration to its retentate concentration — the concentration falls as

$$C/C_0 = \exp(-S \cdot N)$$

so the impurity needs $N = \ln(F)/S_i$ diavolumes for an F-fold clearance while the product is retained as $\exp(-S_p N)$. **A membrane is therefore specified by two coefficients, not one**, and they compound badly: a membrane that partly retains the impurity needs more diavolumes, and every extra diavolume loses more product. The worst case in problem Z4 keeps under half the product. The remedies are a better-matched membrane, a cascade with permeate recycle, or changing the buffer so the impurity's sieving rises.

The operational cost is buffer and time: seven diavolumes on 20 m³ is 140 m³ of buffer, and at a modest permeate flux that is days, which makes diafiltration the schedule bottleneck in many trains.

Cell disruption

High-pressure homogenisation releases intracellular product approximately first order in the number of passes, $1 - e^{-kn}$, with k set by pressure and organism — Gram-positives and yeasts are far tougher than *E. coli*. Two things make the optimum interior rather than "as many passes as needed": **product damage accumulates per pass**, and **fine debris is generated**, which makes the subsequent clarification worse. With 8% quality loss per pass and k = 0.9, the net optimum is three passes at 73%, and a fifth pass destroys product (problem PR-adjacent, computed in PP10).

The alternatives trade differently. **Bead milling** suits tough cell walls and generates fine debris. **Chemical or enzymatic lysis** is gentle, selective and expensive, and leaves the lysing agent as an impurity. For a polymer such as PHA the choice changes the product itself — hypochlorite digestion is cheap and lowers molecular weight, which for a material is the specification rather than a detail.

Breakthrough	DBC < SBC by the unsaturated mass-transfer zone; quote DBC with its residence time
Diafiltration	$C/C_0 = \exp(-S \cdot N)$; impurity needs ln F/S_i; product retained $\exp(-S_p N)$
Disruption	release $1 - e^{-kn}$; damage per pass makes the optimum interior

What to open a book for. Harrison, Todd, Rudge & Petrides, *Bioseparations Science and Engineering*, the chromatography, membrane and disruption chapters; a vendor's process-development handbook for breakthrough-curve protocols.

Problems PR1 to PR8 — protein products

PR1 · extends chapter T2 · 1.5 h

How much can a cell secrete?

Take 100 Sec translocons per cell at 20 residues per second, a 300-residue 33 kDa product, and 10^9 cells per millilitre.

1. Compute the export rate per cell per hour and per litre of broth at full occupancy.
2. Repeat at 10% and 30% occupancy, and give the titer over a 48-hour run.
3. Published secretion titers reach grams per litre. Is that consistent?
4. What would you measure to find whether translocation is your limit?

Worked solution. 100 × 20 ÷ 300 × 3,600 = **24,000 molecules per cell per hour**, which at 10^9 cells/mL is **1.32 g·L⁻¹·h⁻¹** at full occupancy — **0.13 at 10% and 0.40 at 30%**, giving **6.3 and 19 g/L over 48 hours**. So grams per litre is consistent, but only at high cell density and long runs, and only if a substantial fraction of translocons is working on your product rather than the cell's own envelope proteins.

To test whether translocation limits you: compare the **intracellular precursor** (with signal peptide uncleaved) against the mature secreted form. Accumulating precursor means translocation is saturated, and the fixes are a different signal peptide, lower synthesis rate, or co-expression of translocation components. If precursor does not accumulate and product is low, the limit is upstream — transcription, translation or folding — and secretion engineering will not help.

What it teaches: secretion replaces a volume ceiling with a flux ceiling, and the uncleaved precursor is the measurement that tells you which one you are against.

PR2 · extends chapter T1 · 1.5 h

Inclusion bodies: compute the soluble fraction

Soluble fraction ≈ $1/(1 + (k_a/k_f)[U])$.

1. Tabulate the soluble fraction for k_a/k_f of 0.5, 2 and 10 at [U] of 0.1, 1 and 5 µM.
2. What does halving the synthesis rate do at $k_a/k_f = 2$ and [U] = 5 µM?
3. Rank the five standard interventions by what they change in the equation.
4. When is the right answer to stop fighting and go to inclusion bodies deliberately?

Worked solution. At $k_a/k_f = 2$ the soluble fraction is **83%, 33% and 9%** at the three unfolded pools; at 10 it is 50%, 9% and 2%. Halving the synthesis rate halves [U], taking 5 µM to 2.5 and the soluble fraction from **9% to 17%** — a near-doubling of soluble protein from a weaker promoter, at some cost in total.

By mechanism: **induction level and temperature lower [U]**; **chaperones raise k_f**; **fusion partners lower k_a**; **periplasmic targeting changes the compartment** and so both constants. Go to inclusion bodies deliberately when refolding yield is high, disulfides are few, proteolysis in the cytoplasm is a problem, or when the purity of the washed pellet saves a chromatography step — noting that the regulatory burden of demonstrating correct refolding is real for a therapeutic.

What it teaches: solubility is a kinetic competition with a different reaction order on each side, so the dominant lever is always the unfolded pool.

PR3 · extends chapter T2 · 2 h

Refolding: the trade with no interior optimum

1. With yield = 1/(1 + 8C), tabulate yield and protein recovered per litre at 0.01 to 1.0 g/L.
2. Is there an optimum concentration? Explain the shape.
3. Compute the total recovered when 1 g is added in 1, 5 and 20 pulses.
4. Name three process formats that exploit this, and the cost of each.

Worked solution. Yield falls from **93% at 0.01 g/L to 11% at 1.0 g/L**, while protein per litre rises from 0.009 to 0.111 g — so **there is no interior optimum**: yield wants dilution and vessel economics want concentration, and the two never cross. The resolution is to separate local from global concentration. Adding 1 g in **20 pulses at 0.05 g/L each recovers 0.714 g (71%)** against 0.111 g (11%) in a single shot.

Three formats exploit it: **pulse or fed-batch refolding**, cheap and slow; **continuous dilution** into a stirred vessel, which needs good mixing or the local concentration is not what you think; and **on-column refolding**, where the protein is immobilised so chains cannot meet, which gives the highest yield and the lowest throughput. All three are the same idea: keep the chains apart while they fold.

What it teaches: when yield and throughput pull in opposite directions with no optimum, the answer is usually a change of format rather than a better set-point.

PR4 · extends chapter T2 · 1.5 h

Glycosylation: occupancy, mass and host choice

1. A 50 kDa protein has two N-glycan sites, each adding about 2 kDa. What fraction of the molecule is carbohydrate?
2. At 85% occupancy per site, what fraction of molecules is fully glycosylated at 2, 3 and 5 sites?
3. Compute the glycan mass for 10, 50 and 150 mannose residues per site.
4. Choose a host for a therapeutic glycoprotein and justify it from these numbers.

Worked solution. Two sites at 2 kDa on a 50 kDa protein is **7% carbohydrate**. At 85% occupancy the fully glycosylated fraction is **72% at two sites, 61% at three and 44% at five** — so a multi-sequon protein is intrinsically a mixture, and the product specification must describe a distribution rather than a molecule.

Glycan mass: 10 mannose is **1.6 kDa** per site, 50 is **8.1**, and 150 is **24.3** — so yeast hypermannosylation can add half the protein's own mass again, changing both its chromatography and its clearance. For a therapeutic glycoprotein: **bacteria are excluded** (no glycosylation at all), wild-type **yeast is excluded** on immunogenicity and mass, and the choice is between a **humanised yeast** — faster and cheaper, with each pathway edit costing growth — and **CHO**, slower and dearer with acceptable glycans and a regulatory precedent. If the glycan is dispensable for function, the whole argument collapses and *E. coli* returns, which is why that question is asked first.

What it teaches: glycosylation turns "the product" into a distribution, and it is usually the row that decides the host table.

PR5 · extends chapter T1 · 1 h

What a protein product costs the proteome

1. At 30 gDW/L, express titers of 1, 5 and 20 g/L as product per gDW and as a fraction of the 0.5 g/gDW proteome.
2. How is more than one proteome-equivalent possible?
3. Compute the product synthesis rate at 5 g/L over 48 h and compare with total protein synthesis at $\mu = 0.1\ h^{-1}$.
4. What does that ratio imply about where to look for the limit?

Worked solution. 1, 5 and 20 g/L at 30 gDW/L are **0.03, 0.17 and 0.67 g per gDW**, which is **7%, 33% and 133%** of one proteome-equivalent. More than 100% is possible only because the product *leaves* and the cells keep growing — the cell never holds that much at once, it accumulates in the broth. For an intracellular product the same titer would be impossible, which is the whole secretion argument in one number.

The rate tells a different story: 5 g/L over 48 h at 30 gDW/L is $0.003\ g \cdot gDW^{-1} \cdot h^{-1}$ against total protein synthesis of 0.050 at $\mu = 0.1\ h^{-1}$ — the product is **7% of protein synthesis**. So at this titer the ribosome is not the limit, and looking for a limit in transcription, folding or secretion is better founded than raising promoter strength. At 20 g/L the ratio reaches 27% and the proteome argument becomes real.

What it teaches: the same titer is a trivial or an impossible proteome demand depending on whether the product is secreted, and the synthesis-rate ratio says whether expression is even the right thing to push.

PR6 · extends chapter T1 · 1 h

Is chaperone co-expression worth its proteome?

1. Using $\mu/\mu_0 = 1 - \varphi/0.55$, compute the net benefit of chaperones at 2%, 5% and 10% of the proteome for solubility gains of 1.5×, 2× and 3×.
2. What is the break-even gain at a 10% load?
3. Why is the net gain not simply the solubility gain?
4. What experiment establishes the actual gain and the actual load?

Worked solution. Net benefit is the gain times $(1 - \varphi/0.55)$. At a 2% load a 1.5× gain nets **1.45×** and a 3× gain nets

2.89×; at 10% those become **1.23× and 2.45×**. The break-even gain at a 10% load is **1.22×** — so a chaperone set that improves solubility by only 20% at a 10% proteome cost is not worth expressing.

The net is not the raw gain because the chaperones displace the growth-supporting proteome, so you get more soluble protein per cell and fewer cells. The experiment that settles it is two measurements, not one: **soluble active product per litre with and without the chaperones** — which captures both effects at once — and the chaperone fraction of total protein by gel or proteomics, which tells you whether the load is 2% or 10%. Reporting a solubility fold-change from a gel of the soluble fraction alone, as is common, measures the numerator and ignores the denominator.

What it teaches: any co-expressed helper has to clear a break-even set by its own proteome cost, and the right readout is product per litre rather than solubility.

PR7 · extends chapter T2 · 1.5 h

Cytoplasm, periplasm or medium: three ceilings

1. At 2 mL cell volume per gDCW and a periplasm that is 15% of it, compute the periplasmic volume per litre of broth at 30 and 100 gDCW/L.
2. At 5 mM protein in the periplasm (33 kDa), what broth titer is that?
3. Compute the cytoplasmic ceiling at 20% of a 0.5 g/gDW proteome and 100 gDCW/L.
4. Rank the three compartments and say what each costs.

Worked solution. Periplasm is **9.0 mL per litre of broth at 30 gDCW/L and 30 mL at 100**; at 5 mM and 33 kDa that is **1.5 g/L and 5.0 g/L** of broth. The cytoplasmic route at 20% of the proteome and 100 gDCW/L gives **10 g/L**. Secretion into the medium has no intracellular ceiling at all.

So the ranking by ceiling is **medium > cytoplasm > periplasm**, and by ease of purification it is the reverse — the periplasm is a small, relatively clean compartment released by osmotic shock, the cytoplasm needs full lysis and gives you everything, and the medium is the cleanest of all but dilute. The periplasm's compensation is chemistry rather than capacity: it is oxidising, so disulfide-bonded proteins fold there and not in the cytoplasm. That is why a small disulfide-bonded therapeutic goes periplasmic despite the worst ceiling of the three.

What it teaches: compartment choice trades a volumetric ceiling against purification cleanliness and redox chemistry, and the smallest compartment is sometimes the right one.

PR8 · extends chapter T2 · 1 h

Formulation: why titer and concentration are different targets

1. With second-order aggregation, $1/C - 1/C_0 = kt$, compute the time to lose 10% at k of 10^{-4} and 10^{-3} $L \cdot g^{-1} \cdot day^{-1}$ for C_0 of 1 and 10 g/L.
2. State the scaling in one sentence.
3. Why are formulation screens run at the final concentration?
4. What does this mean for a process optimised for titer?

Worked solution. At $k = 10^{-4}$, 1 g/L loses 10% in **1,111 days** and 10 g/L in **111**; at $k = 10^{-3}$ those become 111 and **11 days**. The scaling is simple and brutal: **tenfold the concentration, tenfold faster loss**, because the rate is second order while the loss is measured as a fraction.

Hence screening at the final concentration: a buffer and excipient set that looks stable at 1 g/L can fail at 50 g/L, and the aggregation pathway itself can change — a protein that loses a few per cent as dimer at low concentration may form visible particulates at high. For the process, this decouples two targets that are easy to confuse. **Fermentation titer buys separation cost** (problem PP12); **formulation concentration is set by the dose and the delivery route**, and a high-titer process still has to be concentrated or diluted to that number. Optimising fermentation titer does not optimise the product, and the formulation concentration should be known before the downstream train is designed, because it decides the final ultrafiltration step.

What it teaches: stability scales against concentration, so the final concentration is a specification set outside the process and imposed on it.

Closing notes to part 19

Answering the second question directly

No, the module did not cover the theory behind all the problems — I wrote problems faster than chapters. The audit found **five topics absent** (ANOVA, variance components, equivalence testing, gas holdup and flooding, dissolved CO_2) and **three thin** (response surfaces, diafiltration, protein production). Chapters T1 to T5 close all eight. If you find another problem whose theory is not in a chapter, that is a reasonable thing to send back — the audit was keyword-based and will have missed cases.

What the five chapters add

Chapter	Covers	Serves problems
T1 Protein products I	Proteome limit, folding competition, inclusion bodies, chaperone economics	PR2, PR5, PR6
T2 Protein products II	Sec and Tat capacity, periplasm, refolding, glycosylation, formulation	PR1, PR3, PR4, PR7, PR8
T3 Statistics	ANOVA and interaction, response surfaces, variance components, equivalence	X13, X14, X15, X16
T4 Reactor physics	Heat duty, Kolmogorov shear, holdup and flooding, dissolved CO_2	X4, X9, X10, X11, X12
T5 Downstream	Breakthrough and the mass-transfer zone, diafiltration sieving, disruption	Z1, Z4, PP10

The protein results worth remembering

- **PR1** — translocation capacity is about 1 $g \cdot L^{-1} \cdot h^{-1}$ at full occupancy, so grams per litre needs high density and a long run; accumulating precursor is the diagnostic.
- **PR2** — halving the synthesis rate takes the soluble fraction from 9% to 17%, because aggregation is second order and folding first.
- **PR3** — refolding 1 g in twenty pulses recovers 71% against 11% in one shot.
- **PR4** — at 85% occupancy only 44% of a five-site protein is fully glycosylated; yeast hypermannosylation adds up to 24 kDa per site.
- **PR5** — 20 g/L secreted is 133% of one proteome-equivalent, possible only because it leaves.
- **PR6** — chaperones at a 10% proteome load must deliver 1.22-fold just to break even.
- **PR7** — the periplasm holds only 1.5 g/L of broth at 30 gDCW/L, and is still the right compartment for a disulfide-bonded protein.
- **PR8** — tenfold the formulation concentration, tenfold faster aggregation.

On the numbers

Everything computed here is arithmetic from the stated inputs. Four inputs are recalled rather than computed and are marked in the text: the translocon count and threading rate, the periplasmic volume fraction, the 460 kJ per mole of oxygen, and the glycan masses. They belong on sheet 10 of the lookup workbook. The folding, refolding and aggregation rate constants are illustrative parameters chosen to show the shape of each relation — the *shapes* are the content, the *values* are yours to measure.

Where the set stands

Source	Problems
Your curriculum appendix A1–W3	54
Part 15 X1–X24, Y1–Y6	30
Part 16 Z1–Z9	9
Part 17 MC1–MC5, R1–R5	10
Part 18 PP1–PP12	12
Part 19 PR1–PR8	8
Total	**123**

With G2 from the addendum and the module's own two construction exercises, the tracker counts 125.